MID-SEMESTER RESEARCH PROJECT MILESTONE | 14 October 2026 | Uganda Christian University

Okado Aaron Obbo	Reg No: **S24B23/015** | Student No: B29782	BSc Computer Science

Title: Risk-Adaptive Zero Trust for Securing Hybrid Workforces Using Self-Supervised Anomaly Detection

Problem and Research Objective

Hybrid workforces expose enterprise networks to ransomware, insider attacks, lateral movement and zero-day exploits that legacy VPNs cannot address. ZTA-only systems enforce access control but remain blind to traffic-level anomalies. Neither detects unknown threats without pre-labelled attack data.

Objective: Design and evaluate a Risk-Adaptive ZTA integrating (1) a supervised CNN-GRU for known attack classification and (2) a self-supervised autoencoder for zero-day anomaly detection — compared against three baselines in a controlled four-way evaluation.

Existing Research and Gap

Gambo & Almulhem (2026) reviewed ZTA systematically — strong access control, but no real-time traffic anomaly detection. Bakhshi & Ghita (2021) showed deep learning detects anomalies in encrypted traffic but requires labelled data, failing on novel attacks. Abdelhady et al. (2026) combined lightweight ZTA with ML for IoT but did not address zero-day detection or hybrid workforces. **Gap:** No existing work pairs ZTA with a self-supervised (label-free) anomaly detector in a comparative, hybrid workforce study.

Proposed Contribution

- **Dual-layer ML:** CNN-GRU (known attacks) + Autoencoder (zero-day) — combined into one risk score. Neither alone achieves both.
- **Self-supervised anomaly detection:** Autoencoder trained on NORMAL traffic only. Threshold sensitivity and zero-day separation quality are under investigation — not just accuracy reporting.
- **Four-way controlled evaluation:** Same test set across all systems. Evidence includes FPR, response time and zero-day capability — not accuracy alone.

Full Technical Pipeline / Architecture

INPUT: Live network traffic | NSL-KDD 125,973 train / 22,543 test | Zeek conn.log (Docker) | pcap files

↓

PREPROCESS (src/preprocess.py) — Drop zero-variance feature | log1p outlier scaling | MinMax | 40 clean features

↓

CNN-GRU (Supervised)	Autoencoder (Self-Supervised ★)
Conv1D(64→128→256) + GRU(128→64) 327K params | Test acc 76% | Val acc 98.6% Classes: Normal / DoS / Probe / R2L / U2R	Encoder 40→128→64→16 Decoder 16→64→128→40 Trains on NORMAL only — NO attack labels Anomaly = recon. error > 95th percentile
Known attack classifier	**Zero-day anomaly detector [NEW]**

↓

COMBINED RISK SCORE (0–1) → Flask API → ZTA Policy Engine

↓

>0.8 BLOCK	0.5–0.8 CHALLENGE MFA	0.3–0.5 MONITOR	<0.3 ALLOW

↓

EVALUATION: 4-way comparison (Legacy VPN vs ZTA Only vs ZTA+CNN-GRU vs ZTA+Autoencoder) Metrics: Accuracy | F1 Score | FPR | Response Time | Zero-Day Detection

Progress to Date

Completed	Evidence
NSL-KDD dataset: cleaned, 40 features (noise removed)	src/preprocess.py | scaler.pkl | n_features.pkl
CNN-GRU trained: 50 epochs — test acc 76%, val 98.6%	data/best_model.keras (3.84 MB) | training_results.json
Flask API live: /predict, /health, dashboard	src/api.py | confirmed working on port 5000
Zeek Docker pipeline: pcap → conn.log → API → audit	data/logs/conn.log | data/audit.jsonl
4-way eval framework built; CNN-GRU uses real predictions	src/evaluate.py | data/results/comparison_results.csv
GitHub: 7 commits on branch Aaron	github.com/asdcaaron/zta-ml-framework
Autoencoder architecture built — training PENDING	src/autoencoder.py | autoencoder_model.keras NOT YET generated

Next Six Weeks and Expected Prototype

Wk	Activity	Deliverable
W1	Train autoencoder on normal traffic; validate 95th-pct threshold	autoencoder_model.keras | error distribution plot
W2	Full 4-way evaluation with real autoencoder; compare all systems	Final comparison_results.csv | confusion matrices
W3	GNS3 lab: hybrid topology; live Zeek capture on simulated attacks	Live pipeline demo with real attack traffic
W4	Threshold sensitivity; zero-day recall on unseen attack types	Robustness report | Jupyter notebook
W5	Finalise results; write evaluation chapter; draft poster	Research poster draft | all figures
W6	Complete prototype, poster, submission; record demo	Working prototype + poster submitted

Prototype: Live pipeline — Zeek (Docker) captures traffic, forwards 40 features to Flask API, CNN-GRU + Autoencoder produce a combined risk score, ZTA policy engine enforces ALLOW/MONITOR/CHALLENGE/BLOCK, every decision logged to audit.jsonl. Demonstrable in GNS3 with real simulated attack traffic.

Key References

[1] Gambo & Almulhem (2026). Zero Trust Architecture: A Systematic Literature Review. *Journal of Network and Systems Management*, 34(1). [2] Bakhshi & Ghita (2021). Anomaly Detection in Encrypted Internet Traffic Using Hybrid Deep Learning. *Security and Communication Networks*. [3] Abdelhady et al. (2026). Hybrid ML Anomaly Detection and Lightweight Zero Trust Authentication for LoRaWAN Networks. *IEEE Access*, 14, 18387–18407.